

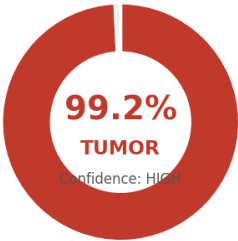
PATHOLOGY AI ANALYSIS REPORT

Computational Pathology Lab | 2026-04-09 14:58 | WSI: camelyon16_test_001

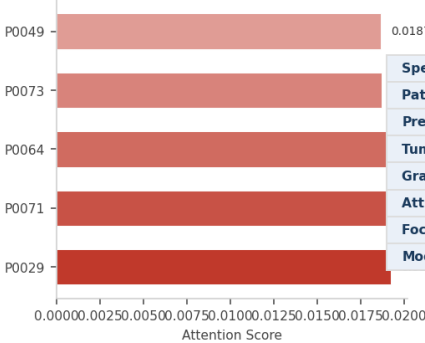
RESULT: TUMOR | Tumor Probability: 99.2% | Confidence: HIGH

AI ANALYSIS: TUMOR (HIGH confidence) | Tumor Probability: 99.2%

Tumor Probability



Top Suspicious Regions
(Attention Score)

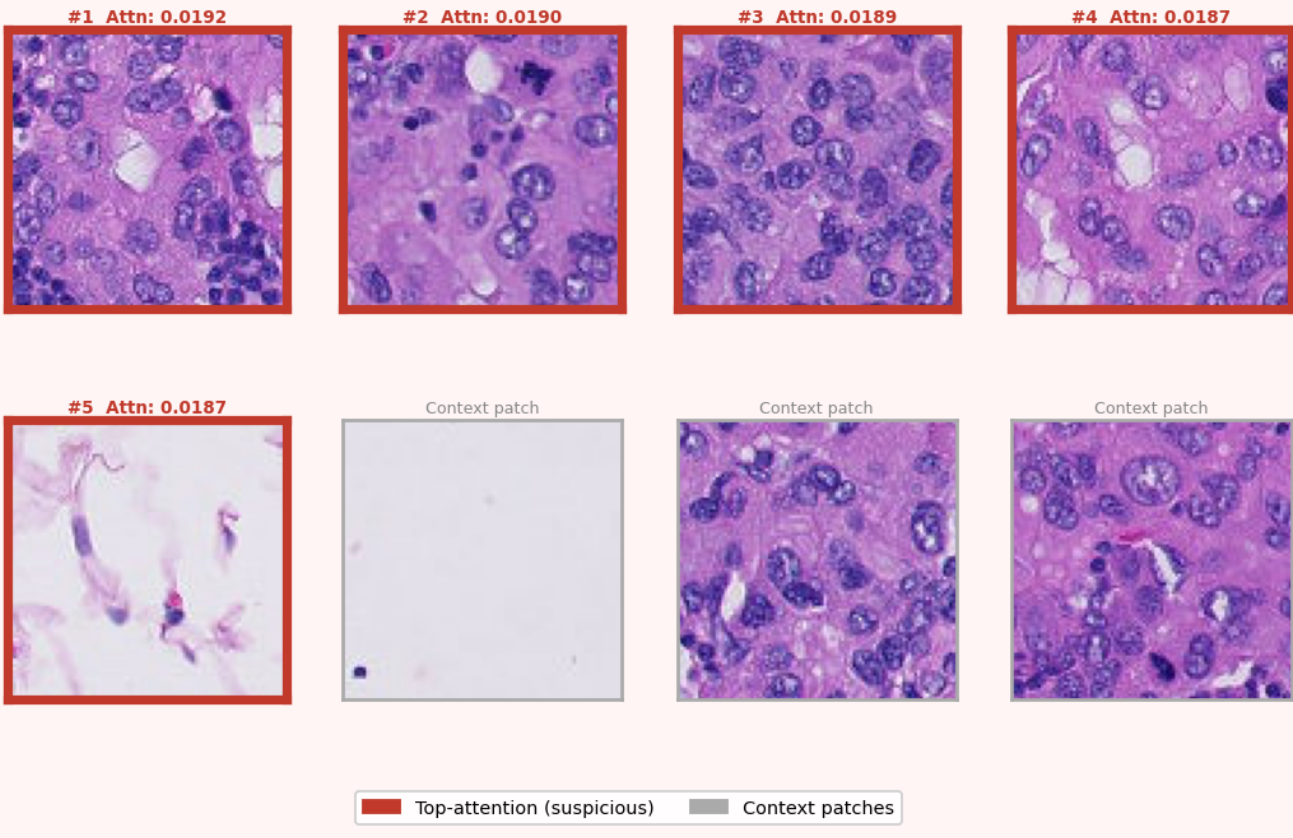


Analysis Details

	Value
Specimen ID	test_001
Patches Analyzed	104
Prediction	TUMOR (HIGH)
Tumor Prob.	99.2%
Grad-CAM IoU	0.413
Attn Entropy	4.281
Focus Pattern	Distributed
Model	EfficientNet-B3 + Gated MIL

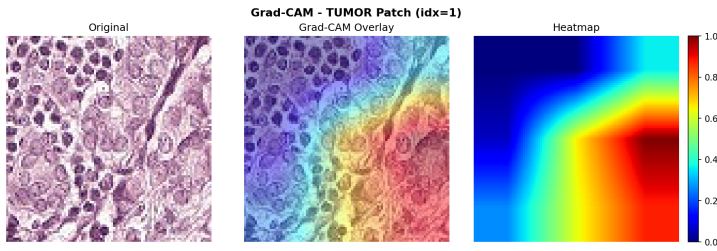
Actual H&E Tissue Patches - Top Attention Regions from This Slide

Test Slide #001 - TUMOR - Top Attention Patches (Real H&E Tissue)



Actual tissue patches extracted from this specific slide. Red-bordered patches are the top-attention regions flagged by the Gated MIL model as most suspicious for malignancy. Attention scores are shown above each patch.

Grad-CAM Heatmap Reference (EfficientNet-B3 Activation)



Grad-CAM activation map highlighting regions of diagnostic relevance as detected by the EfficientNet-B3 backbone. Red=high activation.

1. SPECIMEN INFORMATION

Specimen ID: camelyon16_test_001

Analysis Date: 2026-04-09

Specimen Type: Not specified

Clinical History: Not provided

2. AI ANALYSIS SUMMARY

The specimen was analyzed using the PatchCamelyon MIL - EfficientNet-B3 + Gated Attention model, which examined 104 patches. The model predicted a high probability of tumor presence (99.2%) with a confidence level classified as HIGH. The attention analysis revealed a distributed pattern with multiple suspicious regions identified.

3. DETAILED FINDINGS

The AI model's classification output indicated a strong likelihood of tumor presence, with a tumor probability of 99.2%. The top suspicious regions identified by the model included patches 0029, 0071, 0064, 0073, and 0049. The attention pattern was characterized as distributed, suggesting that the model's attention was spread across multiple regions of the specimen. The attention entropy was calculated to be 4.281. Explainability analysis using Grad-CAM + Gated Attention MIL revealed high activation in lymphocytic clusters with nuclear atypia, with a Grad-CAM IoU of 0.4133.

4. PATHOLOGICAL INTERPRETATION

The AI model's findings suggest a high likelihood of neoplastic processes, potentially indicative of malignancy. However, it is essential to note that these findings are based on computational analysis and require confirmation by a board-certified pathologist. The identification of lymphocytic clusters with nuclear atypia may indicate an immune response or a potential tumor-related phenomenon, but further evaluation is necessary to determine the significance of these findings.

5. RECOMMENDATIONS

Based on the AI model's output, we recommend urgent review of the specimen by a board-certified pathologist to confirm or refute the presence of tumor. Additionally, we suggest performing confirmatory immunohistochemical (IHC) staining, including Ki-67 and HER2, to further characterize the lesion. Correlation with clinical presentation and prior history is also advised. Consideration of a multidisciplinary oncology team consultation may be beneficial for comprehensive patient management.

6. DISCLAIMER

This report is an AI-assisted analysis and should not be considered a definitive diagnosis. The findings and

interpretations presented herein are intended to support and augment the diagnostic process, but human pathologist review and confirmation are essential before any clinical decision-making. The accuracy and reliability of the AI model's output depend on various factors, including the quality of the input data and the limitations of the model itself. Therefore, this report should be used in conjunction with other clinical and pathological evaluations to ensure accurate and reliable diagnosis.

DISCLAIMER: This report is generated by an AI system and is intended solely to ASSIST qualified pathologists. It does NOT constitute a clinical diagnosis. All findings must be reviewed and confirmed by a licensed pathologist. Model: EfficientNet-B3 + Gated Attention MIL. Validated on PatchCamelyon benchmark (AUC > 0.97).